Parametric bootstrap and delta-beta influence for binomial GLMs

The pre-Challenger risk analysis of Dalal, Fowlkes and Hoadley (1989), reproduced.

Tested against	Dalal, Fowlkes and Hoadley (1989), *Journal of the American Statistical Association*: Model (3.2), the bootstrap interval at 30 °F (p. 950, Fig. 5) and the delta beta of each flight (Fig. 7). The 23 flights of Table 1, from the Presidential Commission (1986) (PollisDatasets `challenger`).
Missing in Julia	a parametric bootstrap and case-deletion influence statistics for generalized linear models. GLM.jl fits the models only.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *Parametric bootstrap and delta-beta influence for binomial GLMs* (`glmextras.ipynb`)
Result	every published value: the fit of Model (3.2), the interval at 30 °F, (1.27, 5.95) for "about (1, 6)", and the delta beta of flight 61-A, 4.22 for "about four".

The problem

On the morning of 28 January 1986 the temperature at the launch pad was about 31 °F, far below any earlier launch. Dalal, Fowlkes and Hoadley (1989) asked what the data available before the launch said about O-ring damage at that temperature, and how sure one could be. Two tools answer: a **parametric bootstrap**, for an interval on a prediction far outside the data, where asymptotic formulas are least trustworthy; and **delta beta**, to see whether a single flight drives the fit.

The method

Parametric bootstrap

For a binomial GLM with n trials per case, `parametric` takes the fitted probabilities $\hat{p}$ as true, draws $y^* \sim \text{Binomial}(n, \hat{p})/n$ for every case, refits, and repeats B times Efron (1979). The spread of any refitted quantity, a coefficient or a prediction, estimates its sampling distribution. A sample whose fit fails, as one with all successes or none can, is skipped and counted.

Delta beta

Case i's influence on coefficient j is the change when the case is dropped and the model refitted: $\delta_i = (\hat{\beta}_j - \hat{\beta}_{j(-i)})/S$, with S the standard deviation of the n refitted coefficients (p. 950). Unlike the one-step approximation of Pregibon (1981), it refits exactly, which matters when a few cases carry the fit.

What the tests show

Test	What is compared	Agreement
Model (3.2), p. 948	coefficients, standard errors, G^2	every printed digit
Fig. 5, p. 950	90% interval for damaged O-rings at 30 °F	(1.27, 5.95) for "about (1, 6)"
Fig. 7, p. 950	the most influential flight	61-A, 4.22 s.d. for "about four"
Fig. 7, p. 950	the next three	flights 2, 11 and 14, as published

The tests are coarse because the paper's are: it prints the interval and the delta beta to no decimals and shows the rest in figures. They confirm that the methods do what the paper did, not their last digits. The bootstrap is random: its interval moves in the second decimal with the seed (1,000 samples are drawn).

What the replication adds

The paper's "about (1, 6)" becomes a number: of six field-joint O-rings, between 1.27 and 5.95 expected damaged at 30 °F. And the influence of flight 61-A can be asked directly: dropped, its damage at a warm 75 °F no longer softens the fit, the model expects nearly all six O-rings to be damaged, and the interval narrows to the top of the scale. The notebook's example shows both intervals.

Still open

The paper also bootstraps a model of primary erosion with temperature and pressure (p. 953). About a third of its samples have no maximum likelihood estimate: one coefficient runs off to infinity (separation), and the paper does not say what its

program did with them. Those intervals are not used as tests; the Logistic Regression workflow shows them.

Further considerations

- **Separation.** The failed samples of the pressure model are a known weakness of maximum likelihood with few events. The penalised likelihood of Firth (1993) would give every sample a finite estimate; it belongs here only once a published table tests it.
- **Beyond binomial.** Both functions need only a family's random draw and a refit, so Poisson and gamma GLMs follow; each needs its own published test.
- **Upstream.** Both are a few lines over GLM.jl and could be offered to it, or to a GLM diagnostics package, with the Challenger values as tests.

References

Dalal, S. R., Fowlkes, E. B. and Hoadley, B. (1989). Risk analysis of the space shuttle: pre-Challenger prediction of failure. *Journal of the American Statistical Association* 84(408), 945–957. doi:10.1080/01621459.1989.10478858

Efron, B. (1979). Bootstrap methods: another look at the jackknife. *The Annals of Statistics* 7(1), 1–26.

Firth, D. (1993). Bias reduction of maximum likelihood estimates. *Biometrika* 80(1), 27–38.

Pregibon, D. (1981). Logistic regression diagnostics. *The Annals of Statistics* 9(4), 705–724.

Presidential Commission on the Space Shuttle Challenger Accident (1986). *Report to the President*, Vol. 1, pp. 129–131: the data.

Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.

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